

Introduction aux ontologies biomédicales

École d'été interdisciplinaire en numérique de la santé (EINS 2026)

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Plan

- **Partie I** : Les ontologies comme solution au problème de Babel de l'informatique médicale
- **Partie II** : Les différents niveaux d'ontologies
- **Partie III** : Définitions textuelles et formelles
Application à la classification : l'exemple de la maladie



Partie 1 : Les ontologies comme solution au problème de Babel de l'informatique médicale



Les données médicales au Québec aujourd'hui

- Cabinets de médecin
- Hôpitaux
- RAMQ
- Centre Local de Service Communautaire (CLSC)
- Ministère de la Santé et des Services Sociaux (MSSS)
- Cohortes
- Essais cliniques
- Biobanques
- ...



Tour de Babel des systèmes
d'information

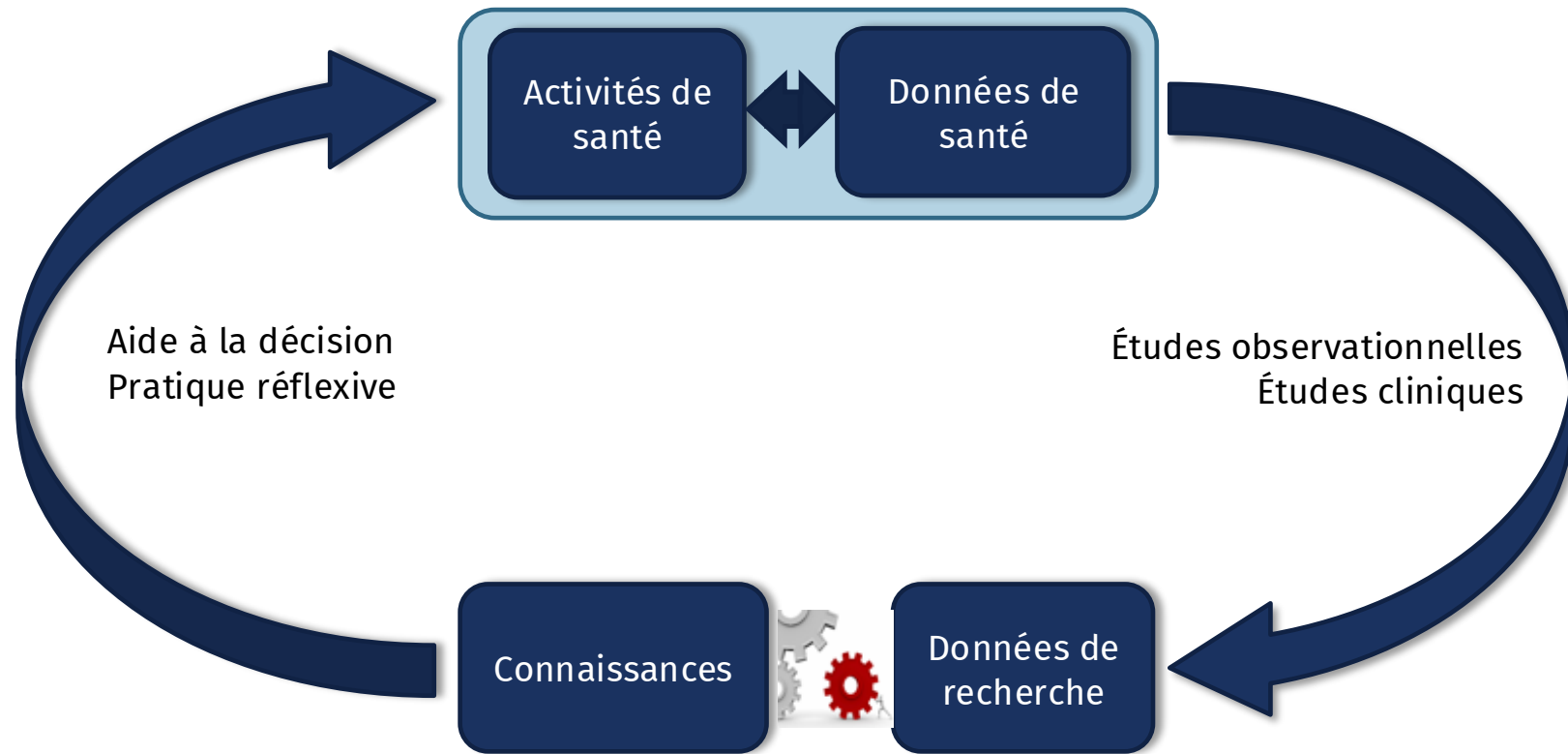
Les données médicales au Québec aujourd'hui

- Défi : rendre l'information scientifique interoperable, partageable, réutilisable
- Obstacle principal : diversité des données, plutôt que leur quantité
- “Problème de Babel” :
 - Idiosyncrasie technologique
 - Idiosyncrasie humaine



Tour de Babel des systèmes d'information

Un défi pour la mise au point de systèmes de santé apprenants



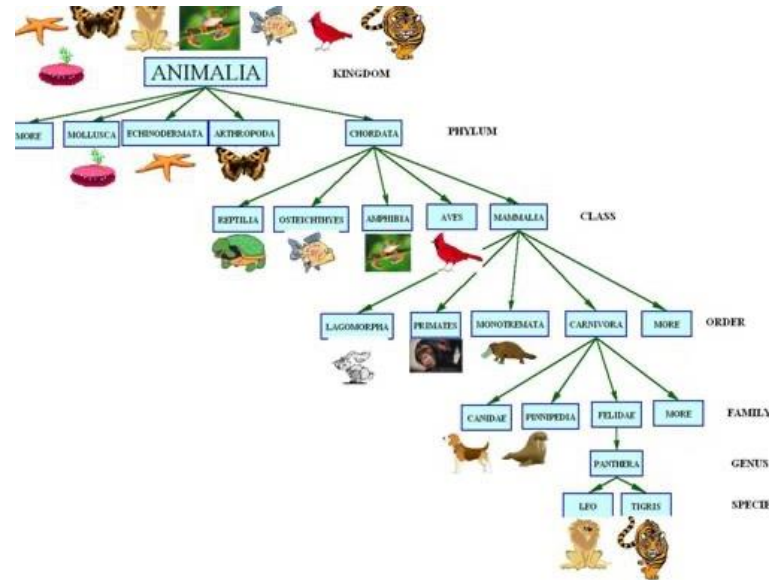
Approche du GRIIS : Utiliser les ontologies



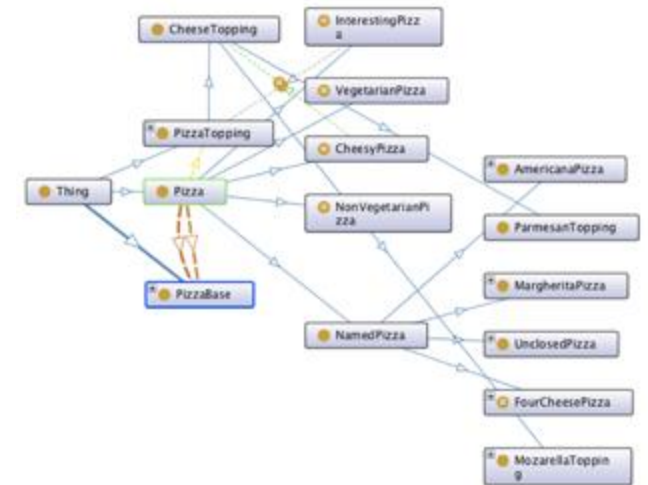
Terminologies, taxonomies et ontologies

Medicines and Drugs	
Term	Definition
abortifacients	used to stimulate uterine contraction and promote evacuation of the uterus to cause abortion
ACE inhibitor	used to block the enzyme responsible for converting Angiotensin 1 to angiotensin 2 in the lungs; this blocking prevents vasoconstriction
antiarrhythmics	this affects the action potential of cardiac cells and are used to treat arrhythmias and return normal rate and rhythm of the heart muscles
anticoagulant	drugs that inhibit any step of the coagulation process, preventing or slowing clot formation

Terminologie



Taxonomie

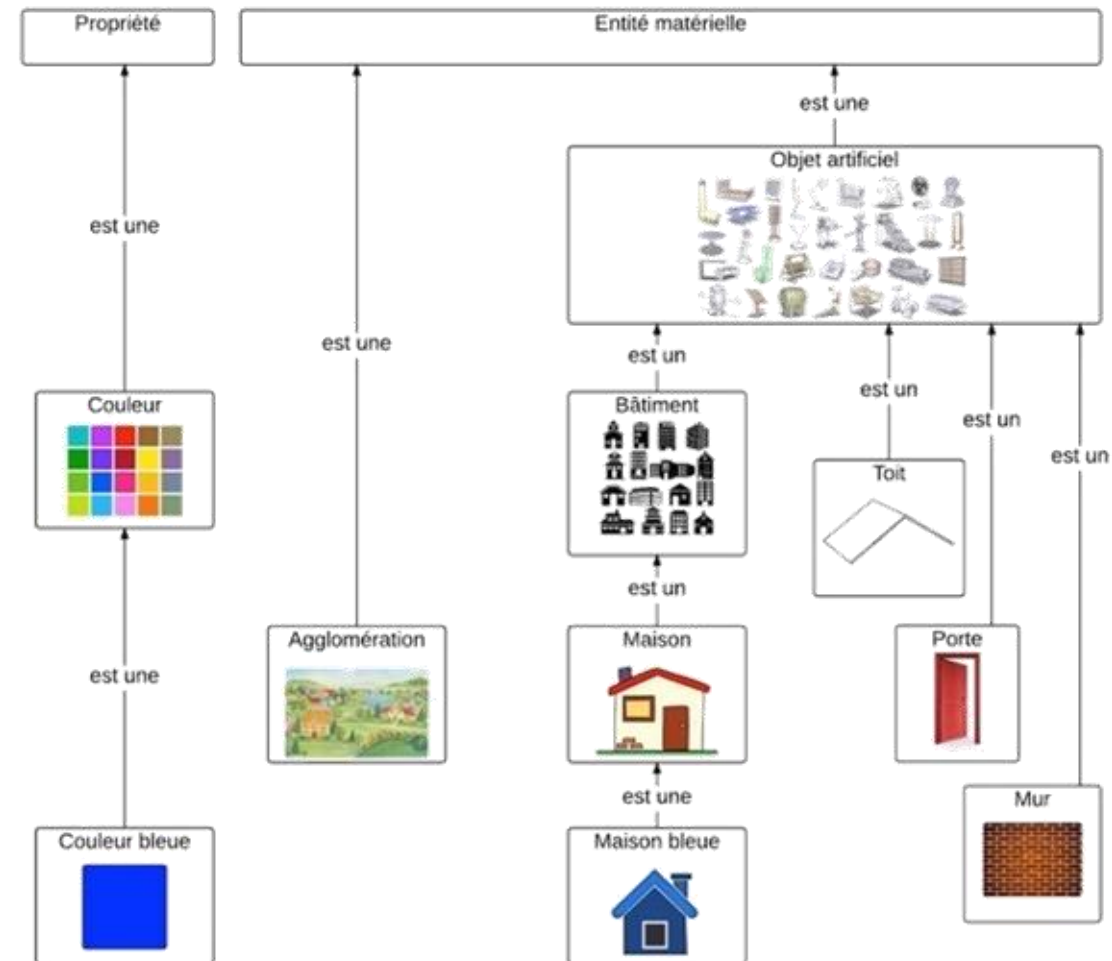


Ontologie

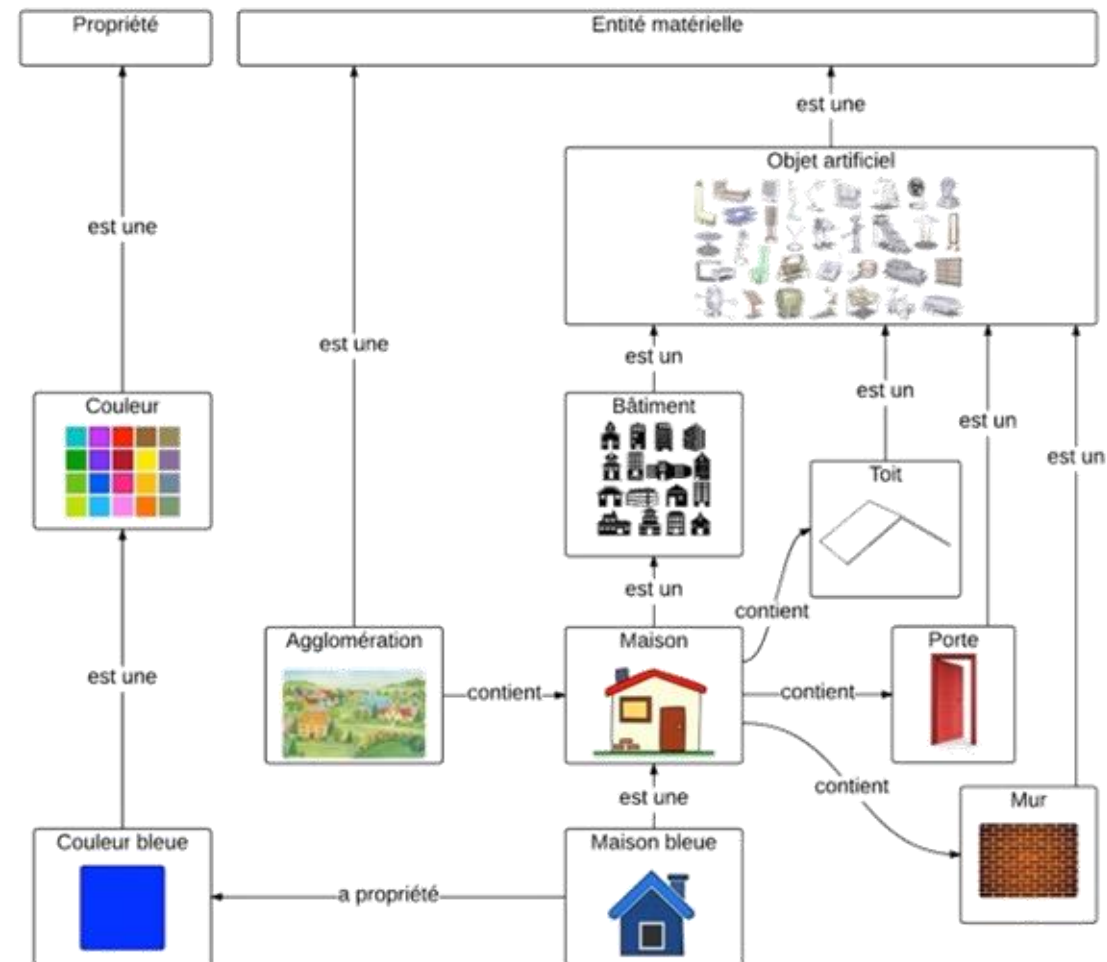
Terminologie

Terme	Définition
Maison	Bâtiment destiné à servir d'habitation à des humains.
Toit	Surface supérieure d'un bâtiment.
Porte	Ouverture limitant un espace clos, permettant la communication entre cet espace et ce qui est extérieur à cet espace.
Mur	Ouvrage de maçonnerie vertical (parfois oblique), élevé sur une certaine longueur pour constituer le côté d'un bâtiment.

Taxonomie



Ontologie



OWL (Web Ontology Language),
fondé sur la logique
de description (DL)

Description: House

SubClass Of +

- **Building**
- **has_part some Door**
- **has_part some Roof**
- **has_part some Wall**

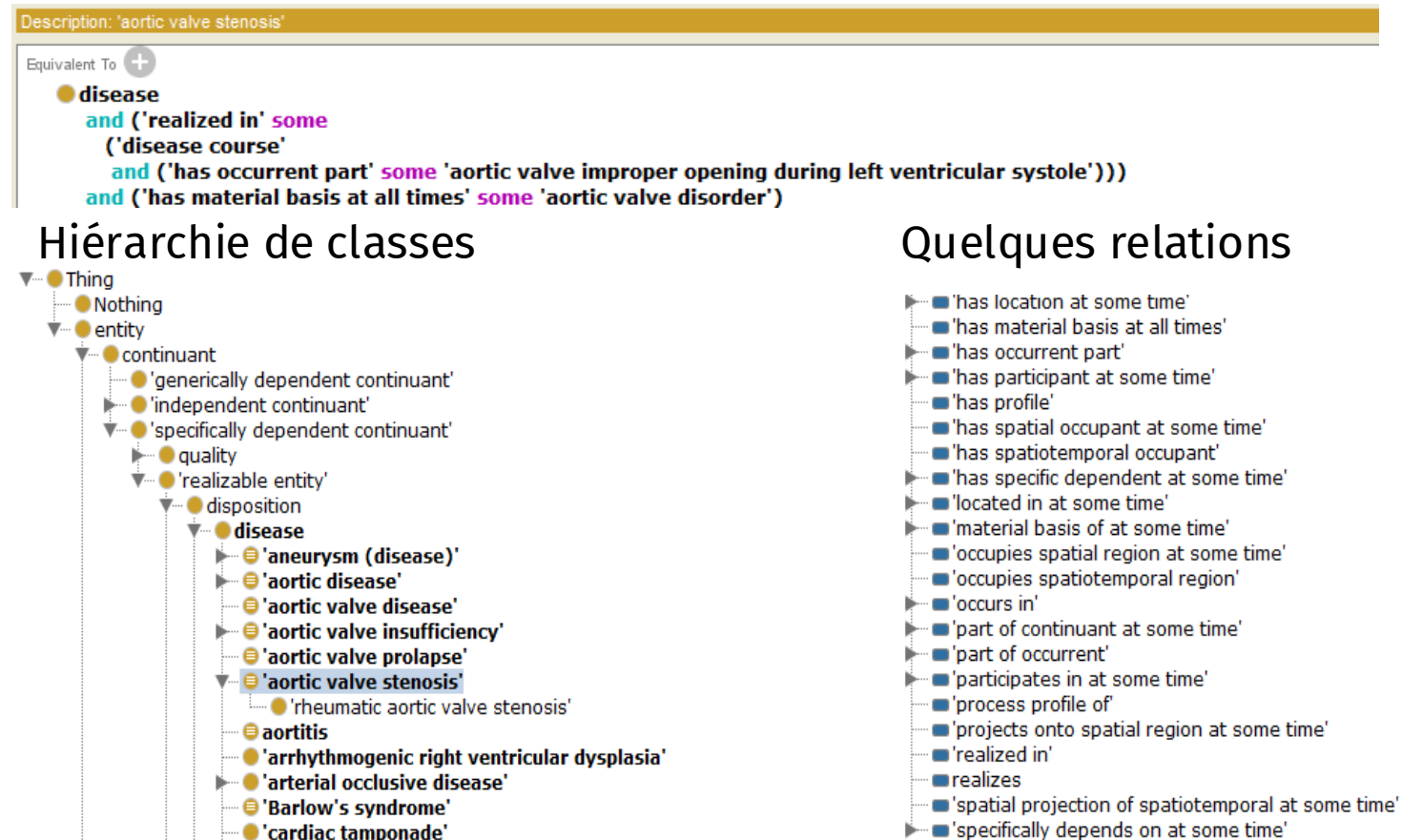
Capture d'écran de Protégé

Ontologie de domaine

Définition formelle de la sténose aortique valvulaire (OWL)

Représentation canonique des entités et relations d'un domaine donné, visant à l'exhaustivité.

Exemple : Cardiovascular Disease Ontology (CVDO)



Vues d'écran du logiciel Protégé

Moteurs de recherche pour ontologies

OLS
ONTOLOGY SEARCH

Home | **Ontologies** | Help | About | Downloads

Show 10 Search ontologies...

Ontology	ID	Description
Alzheimer's Disease Ontology (ADO)	ADO	Alzheimer's Disease Ontology is a knowledge-based ontology that encompasses varieties of concepts related to Alzheimer's Disease, fundamentally structured by upper level Basic Formal Ontology (BFO). This Ontology is enriched by the interrelational entities that demonstrate the nextwork of the understanding on Alzheimer's disease and can be readily applied for text mining.
African Population Ontology	AFPO	AFPO is an ontology that enables annotating African individuals and combines knowledge accumulated about existing populations with their genetic fingerprints in a standardized format. The AFPO can be employed to classify African study participants comprehensively in prospective research studies. It can also be used to classify past study participants by mapping them using a language or ethnicity identifier or synonyms.
Agronomy Ontology	AGRO	AgrO is an ontology for representing agronomic practices, techniques, variables and related entities

Ontobee

Home | Intro | Statistics | SPARQL | Ontobee | Annotator | Tutorial | FAQs | References | Links | Contact | Acknowledge | News

Welcome to Ontobee!

Ontobee: A [linked data](#) server designed for ontologies. Ontobee is aimed to facilitate ontology data sharing, visualization, query, integration, and analysis. Ontobee dynamically [dereferences](#) and presents individual ontology term URIs to (i) [HTML](#) web pages for user-friendly web browsing and navigation, and to (ii) [RDF](#) source code for [Semantic Web](#) applications. Ontobee is the default linked data server for most [OBO Foundry library ontologies](#). Ontobee has also been used for many non-OBO ontologies.

Please select an ontology (optional)

Keywords: Search terms Batch Search

Jump to <http://purl.obolibrary.org/obo/> Go

Currently Ontobee has been applied for the following ontologies:

No.	Ontology Prefix	Ontology Full Name	OBO	List of Terms
1	ADO	Alzheimer's Disease Ontology	L	
2	AEO	Anatomical Entity Ontology	L	
3	AFO	Allotrope Foundation Ontology	N	

Welcome to BioPortal, the world's most comprehensive repository of biomedical ontologies

Search for a class

Enter a class, e.g. Melanoma

[Advanced Search](#)

Find an ontology

Start entering ontology name, e.g. Cancer, then choose from list

[Browse Ontologies](#)

Ontology Visits (February 2018)

More

BioPortal Statistics

Ontologies	692
Classes	8,847,370
Resources Indexed	48
Indexed Records	39,537,360
Direct Annotations	95,468,433,792
Direct Plus Expanded Annotations	144,789,582,932

Les ontologies comme légendes pour les bases de données

MouseEcotope

Tool	Statistical model	Correction for multiple experiments	GO Visualization	Microarrays supported	Time to process 200 genes (s)
Auto-Express	χ^2 , binomial, hypergeometric, Fisher's exact test	Sidak, Holm, Benjamini, FDR	Flat, Tree	172 commercial arrays (Affymetrix, Superscript, Sigma Genosys, Clontech, PerkinElmer, Openix, Takara, NDA), can also upload a user-defined list	7, 8, 16, 28
GoldMine	Fisher's exact test	Relative enrichment	Tree, DAG	uploads from user	77, 123, 223, 34
DAVID	None	None	Not available	Not applicable	13, 17, 27, 54
EASE online	Fisher's exact test	Benjamini	Not available	27 arrays (Affymetrix only), can also upload a user-defined list	15, 18, 34, 74
GeneSinger	Hypergeometric	Benjamini	Flat, no hierarchical structure	Uploads from user	4, 6, 6, 8
FuncAssociate	Fisher's exact test	None	Not available	Uploads from user	22, 27, 29, 50
GOTM	Hypergeometric	None	Tree	37 arrays (Affymetrix only), uploads from user	39, 60, 137, 7
FastGO	Percentage	Step-down method, FDR (Benjamini and Hochberg, 1995), FDR (Benjamini and Yekutieli, 2001)	Flat, Tree	Uploads from user	15, 49, 69, 105
CLENCB	Hypergeometric, χ^2 , binomial	None	DAG	Uploads from user	NA
GOstat	χ^2 , Fisher's exact test	FDR, Holm	Not available	Uploads from user	12, 20, 46, 8
GOToolbox	Hypergeometric, binomial, Fisher's exact test	Benjamini, Holm, Hochberg, Bonferroni, FDR	Not available	Uploads from user	22, 81, 143, 7
GoMiner	χ^2	not available	DAG	23 arrays (Affymetrix only), uploads from user	7, 7, 7, 7
Ontology Transposer	Hypergeometric	FDR	Not available	5 arrays (Affymetrix), uploads from user	NA
eGOs	Binomial	None	Tree	Uploads from user	20, 43, 80, 95

**sphingolipid
transporter
activity**

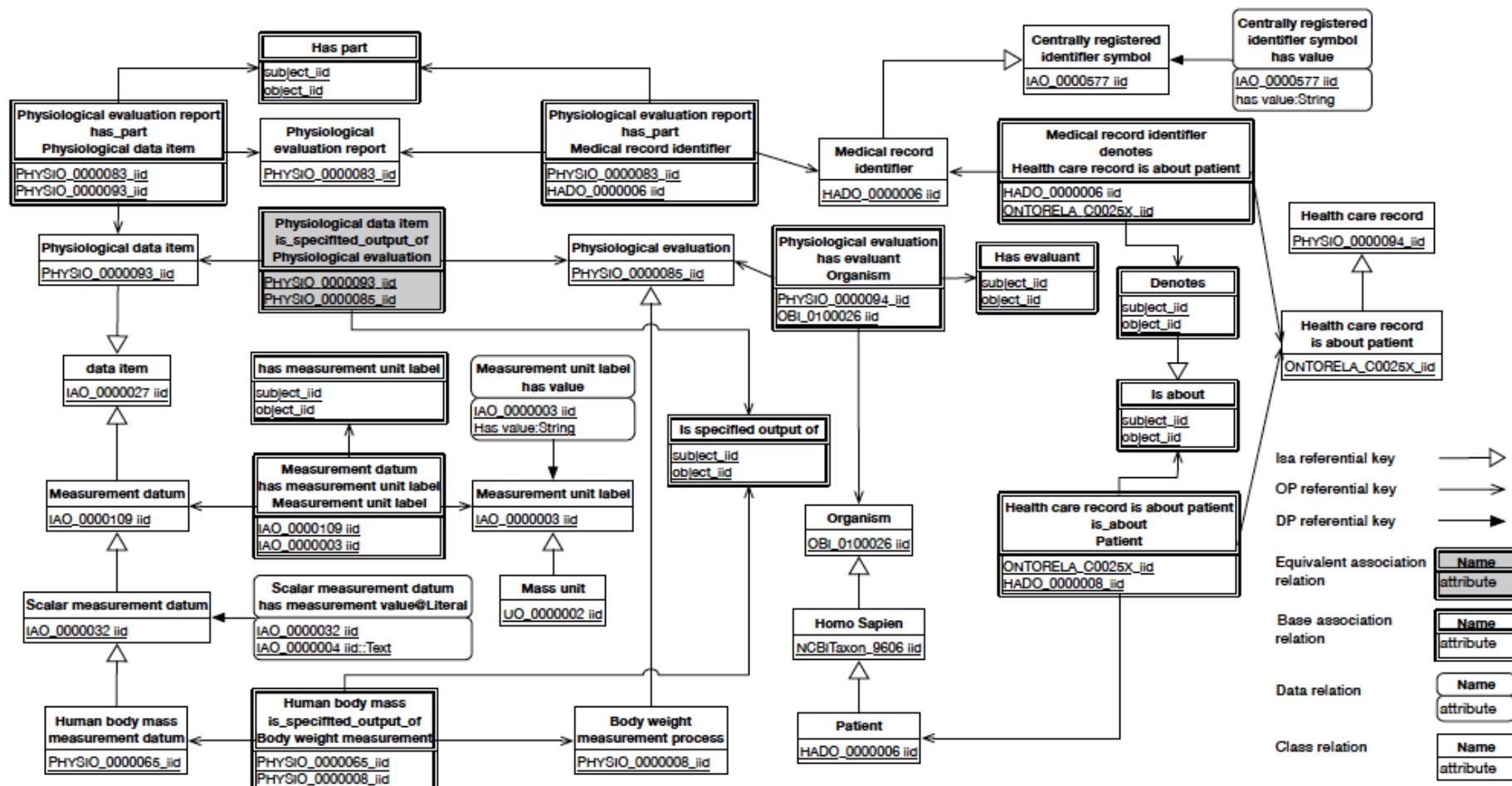
DiabetInGene

GluChem

GlyProt

H4AC Histone-acetate ligase
 HMOX1 Heme oxygenase (decycling) 1
 Hmox-1 Hemoxygenolase Hmox-1
 Erythrocyte anhydrase 50kd glycoprotein
 CAC Carbonic anhydrase II
 NF1B Nuclear factor 1/B
 HMMHCIN PROTEIN Hm-45
 Hemochin Protein 45, Class I
 HMOX2 Heme gene, partial cds
 HMOX2 Histatin 1, neutrophil
 HMO Hemoxygenolase
 CYS2 Cystatin C
 AHRB Rat homolog, member 6 (rho 6)
 HMOX1 Hemoxin 1 (Lipoxygenin 3)
 CCHER
 NF1B Integration oncogene
 P component of complement
 NF-IL6-beta
 HMOX1 Hemoxin-1
 HMOX1 Alpha hemoxygenase II isozyme
 HMOX1 Hemoxygenase
 Hemolys suppressor-of-white-sprint
 CAC antigen (c50)
 HMOX1 HMOX1
 HMOX1B Small proline-rich protein 1B
 ACTN1 Actinin alpha 1
 HMOX1 HMOX1 gene, partial cds
 CTHRE Decarboxylate kinase
 Carboxymethyl antigen precursor
 CTHS Lymphoblastin-beta
 GSHB GSH-binding protein 3
 HS SHP = Escherichia coli unknown
 GSHAPLAIN CYCLININ HMOX1
 Spinal Muscular Atrophy 4
 HMOX1 HMOX1 gene, partial cds
 Erythrocytic (HMF)
 HMOX1 Hemoxygenolase LMOF-H1
 HMOX1 Replication protein 21
 Class II mRNA
 CTHS CTH synthetase
 HMOX1 HMOX1-LIKE PROTEIN HMOX1
 (HMOX1) mRNA
 HMOX1 calcium-binding protein 113
 HMOX1 Protein-tyrosine kinase blot
 TMB (H3)
 HMOX1
 TMB
 CTHS antigen
 Spinal: muscle abundant protein

Utilisation d'une ontologie pour créer un modèle de base de données relationnelles



Khnaissar, Christina, et al. Using an ontology to derive a sharable and interoperable relational data model for heterogeneous healthcare data and various applications. *Methods of Information in Medicine* 61.S 02 (2022): e73-e88.

Partie 2 : Les différents niveaux d'ontologie



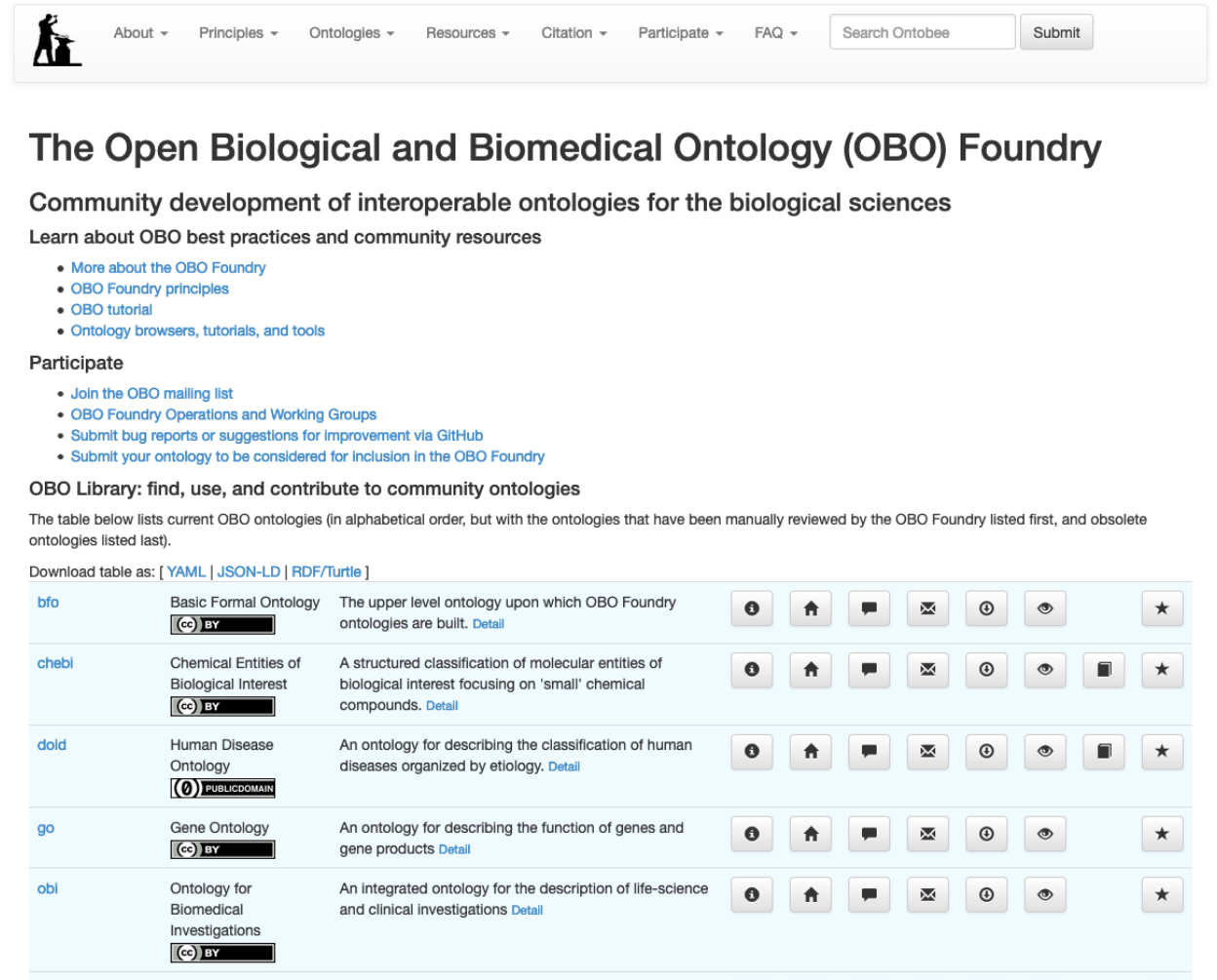
Difficultés

- Formation de silos de données d'un même domaine : la multiplication de terminologies et ontologies recrée le problème de l'interopérabilité à un plus haut niveau.
- La multiplicité des domaines représentés par des ontologies : médecine, biologie, industrie, géographie, éducation...



Solution

- Créer des ensembles d'ontologies de domaine :
- avec une seule ontologie par domaine
- unifiées par une ontologie de haut niveau.



The screenshot shows the OBO Foundry website. At the top is a navigation bar with links: About, Principles, Ontologies, Resources, Citation, Participate, and FAQ. There is also a search bar labeled 'Search Ontobee' and a 'Submit' button. Below the navigation bar is the main heading 'The Open Biological and Biomedical Ontology (OBO) Foundry' followed by the subtitle 'Community development of interoperable ontologies for the biological sciences'. A section titled 'Learn about OBO best practices and community resources' lists links: 'More about the OBO Foundry', 'OBO Foundry principles', 'OBO tutorial', and 'Ontology browsers, tutorials, and tools'. Another section titled 'Participate' lists links: 'Join the OBO mailing list', 'OBO Foundry Operations and Working Groups', 'Submit bug reports or suggestions for improvement via GitHub', and 'Submit your ontology to be considered for inclusion in the OBO Foundry'. Below this is the 'OBO Library: find, use, and contribute to community ontologies' section, which includes a table of current OBO ontologies. The table lists ontologies like bfo, chebi, doid, go, and obi, each with a brief description and a 'Detail' link. At the bottom left of the screenshot are the logos for GRIIS and Université de Sherbrooke.

About ▾ Principles ▾ Ontologies ▾ Resources ▾ Citation ▾ Participate ▾ FAQ ▾ Search Ontobee Submit

The Open Biological and Biomedical Ontology (OBO) Foundry

Community development of interoperable ontologies for the biological sciences

Learn about OBO best practices and community resources

- [More about the OBO Foundry](#)
- [OBO Foundry principles](#)
- [OBO tutorial](#)
- [Ontology browsers, tutorials, and tools](#)

Participate

- [Join the OBO mailing list](#)
- [OBO Foundry Operations and Working Groups](#)
- [Submit bug reports or suggestions for improvement via GitHub](#)
- [Submit your ontology to be considered for inclusion in the OBO Foundry](#)

OBO Library: find, use, and contribute to community ontologies

The table below lists current OBO ontologies (in alphabetical order, but with the ontologies that have been manually reviewed by the OBO Foundry listed first, and obsolete ontologies listed last).

Download table as: [YAML](#) | [JSON-LD](#) | [RDF/Turtle](#)

bfo	Basic Formal Ontology 	The upper level ontology upon which OBO Foundry ontologies are built. Detail	      
chebi	Chemical Entities of Biological Interest 	A structured classification of molecular entities of biological interest focusing on 'small' chemical compounds. Detail	       
doid	Human Disease Ontology 	An ontology for describing the classification of human diseases organized by etiology. Detail	       
go	Gene Ontology 	An ontology for describing the function of genes and gene products Detail	      
obi	Ontology for Biomedical Investigations 	An integrated ontology for the description of life-science and clinical investigations Detail	      

Quatre types d'ontologies

- **Ontologie de haut-niveau** : indépendante du domaine

Exemple: Basic Formal Ontology (BFO): Material entity, Process...

- **Ontologie de niveau intermédiaire**: utilisable par plusieurs ontologies de domaine (mais pas toutes).

Exemple: Ontology for General Medical Science (OGMS): Disease, Pathological process...

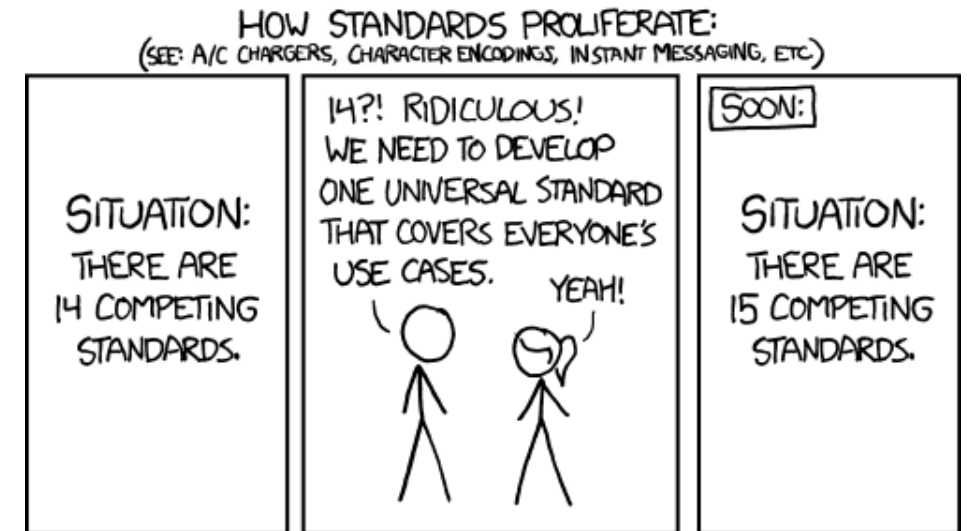
- **Ontologie de domaine** (ou de référence): représentation canonique des entités et relations d'un domaine donné.

Exemple: Cardiovascular Disease Ontology (CVDO): Aortic disease, Atrial fibrillation...

- **Ontologie d'application**: créée pour accomplir certains buts spécifiques.

Ontologies de haut niveau

- Plusieurs ontologies de haut niveau proposées :
 - BFO (Basic Formal Ontology)
 - DOLCE (Descriptive Ontology for Linguistic and Cognitive Engineering)
 - UFO (Unified Foundational Ontology)
 - GFO (General Formal Ontology)
 - ...



<https://xkcd.com/927/>

Contexte interdisciplinaire

- La recherche de cohérence en développement d'ontologie a mené à des collaborations entre :
 - informaticiens et spécialistes des sciences de l'information
 - biologistes et cliniciens
 - mais aussi philosophes, logiciens, et linguistes.

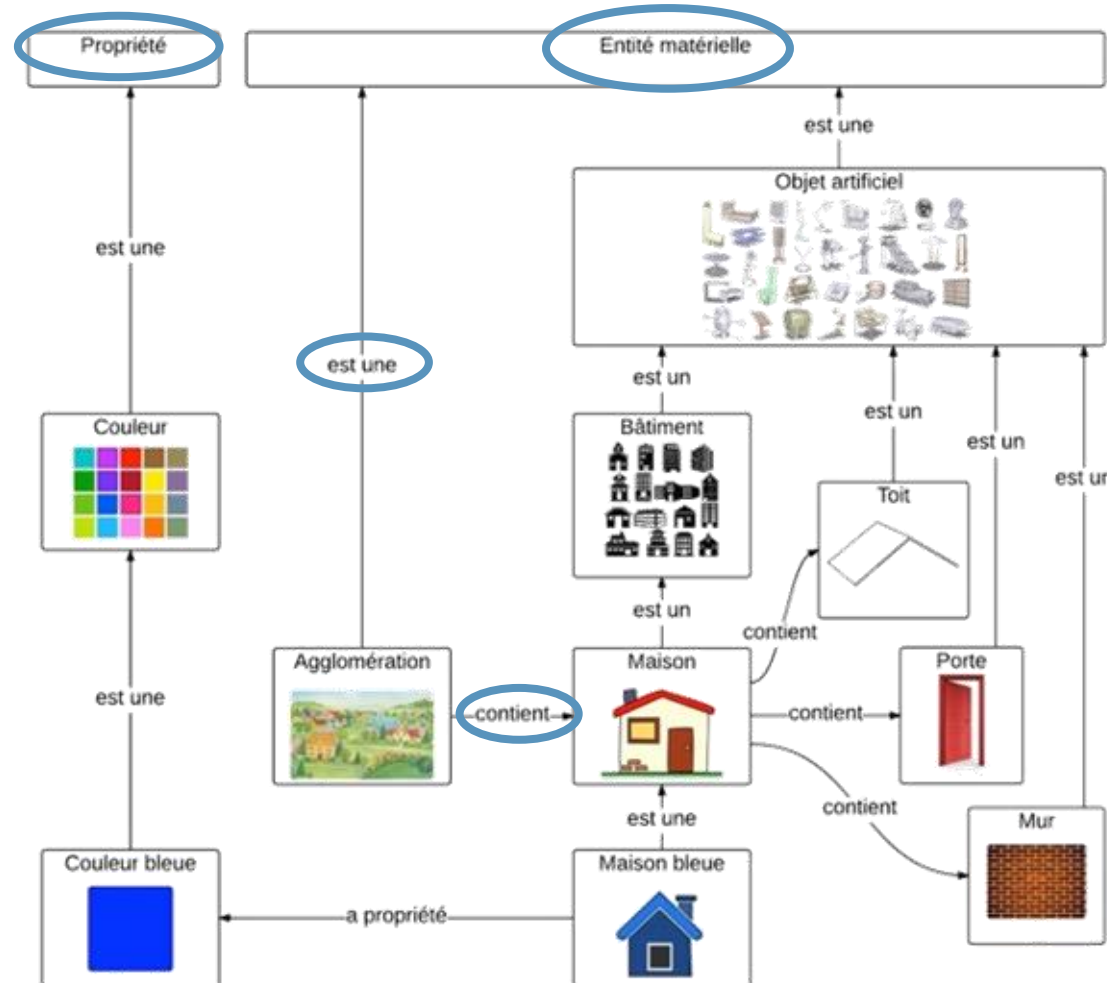


Pourquoi des philosophes ???

L'apport de la philosophie

Apports de la métaphysique :

- similarité
- identité et changement
- causalité
- méréologie (relation de tout à partie)
- ...



Apports de la philosophie des sciences :

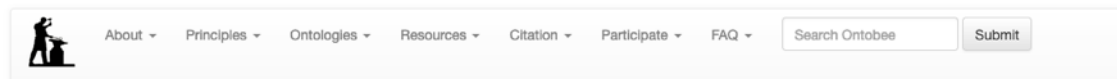
- maladie
- risque
- probabilités
- ...

BFO (Basic Formal Ontology)

- Ontologie à méthodologie dite « réaliste » : vise à décrire le réel en s'alignant sur nos meilleures théories scientifiques.
- Développée par différentes institutions, notamment à l'université de Buffalo
- Standard ISO IEC 21838-2:2021

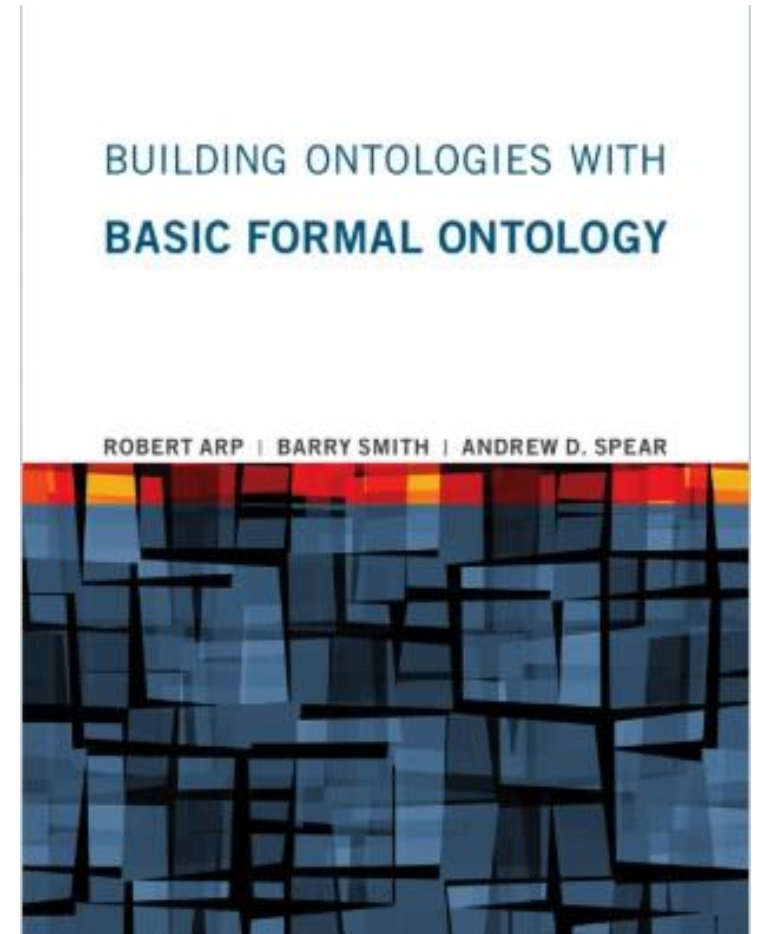
[https://standards.iso.org/ittf/PubliclyAvailableStandards/c074572_ISO_IEC_21838-2_2021\(E\).zip](https://standards.iso.org/ittf/PubliclyAvailableStandards/c074572_ISO_IEC_21838-2_2021(E).zip)

- Ontologie très utilisée dans le domaine biomédical, de plus en plus dans le domaine industriel.
- Largement utilisée par les ontologies de la OBO Foundry



The Open Biological and Biomedical Ontology (OBO) Foundry

Community development of interoperable ontologies for the biological sciences



Continuants et occurrents

Continuants (perspective 3D)



Mitochondrie

Cellule

Organe

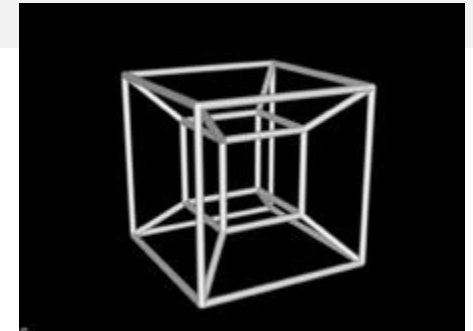
Organisme

Espèce

...

existent
pleinement à
tout instant de
leur existence

Occurrents (perspective 4D)



Production d'ATP

Division cellulaire

Circulation sanguine

Enfance

Spéciation

...

n'existent pas
pleinement à un
instant : ils ont des
parties temporelles

Continuants indépendants et dépendants

Universaux et particuliers

	Continuant indépendant	Continuant dépendant	Occurrent
	<i>Humain</i> <i>Chat</i> <i>Table</i>	<i>Rougeur</i> <i>Forme rectangulaire</i> <i>Fragilité</i>	<i>Conférence</i> <i>Souper</i> <i>Circulation sanguine</i>

Continuants indépendants et dépendants

Universaux et particuliers

	Continuant indépendant	Continuant dépendant	Occurrent
Universaux / Classes	<i>Humain</i> <i>Chat</i> <i>Table</i>	<i>Rougeur</i> <i>Forme rectangulaire</i> <i>Fragilité</i>	<i>Conférence</i> <i>Souper</i> <i>Circulation sanguine</i>
Particuliers	cet humain ce chat cette table	cette rougeur cette forme rectangulaire cette fragilité	cette conférence ce souper cette circulation sanguine

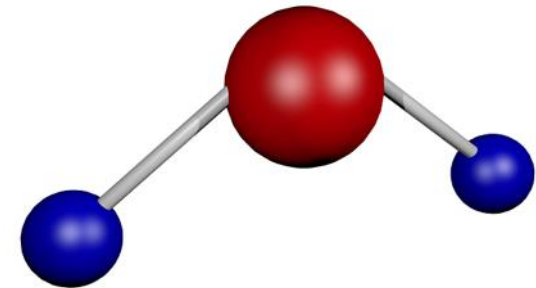
Le besoin de définitions textuelles : Eviter les imprécisions de pensée



Ambiguïté

- La “International Classification of Nursing Practice” (ICNP) définit "water" comme :

”A type of Nursing Phenomenon of Physical Environment with the specific characteristics: clear liquid compound of hydrogen and oxygen that is essential for most plant and animal life influencing life and development of human beings.”



Circularité

First Healthcare Interoperability Resources Specification (FHIR) :

- Nourriture (“food”) défini comme “naturally occurring, processed, or manufactured entities that are primarily used as food for humans and animals.”
- Contenant (“container”) défini comme “a container of other entities”



Confusion “utilisation-mention” (“use-mention”)

- HL7 définit un sujet vivant (“living subject”) comme : “a subtype of Entity representing an organism or complex animal, alive or not.”
- Confond un objet (un sujet vivant) avec sa représentation (“Entity representing [...]).
- E.g. : “Le sommeil est bon pour la santé et contient trois voyelles.”



Un modèle utile : la définition aristotélicienne

- Exemple historique chez Aristote :

Humain =_{def} Un animal qui est rationnel.

- Définition aristotélicienne d'une classe fille ("espèce" E) par sa classe parente ("genre" G), en la particularisant avec un "*differentia*" (D) :

E =_{def} Un G qui D.

Exemple (FMA) : Heart =_{def} An organ with cavitated organ parts, which is continuous with the systemic and pulmonary arterial and venous trees.



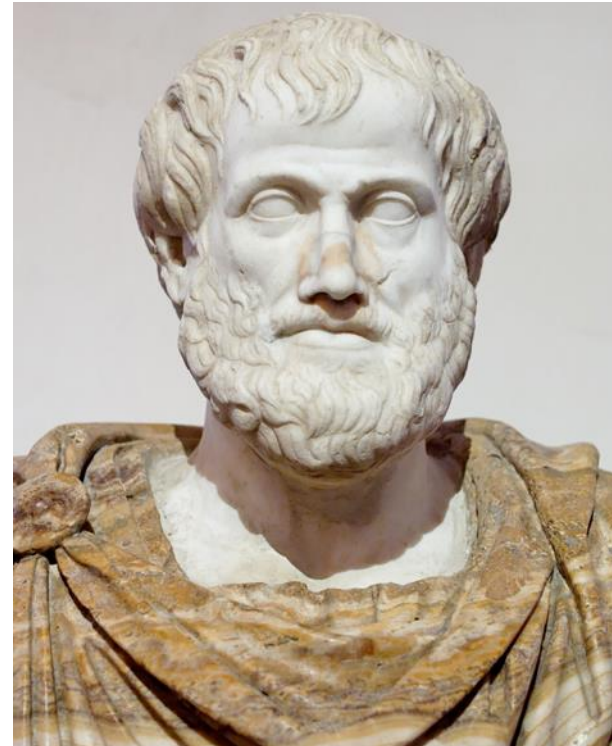
CURRENT ISSUES - PERSPECTIVES AND REVIEWS

Transitive or Not: A Critical Appraisal of Transitive Inference in Animals

David Guez* & Charles Audley†

* School of Psychology, The University of Newcastle, Newcastle, NSW, Australia

† Centre for Applied Psychology, Faculty of Health, University of Canberra, Canberra, ACT, Australia

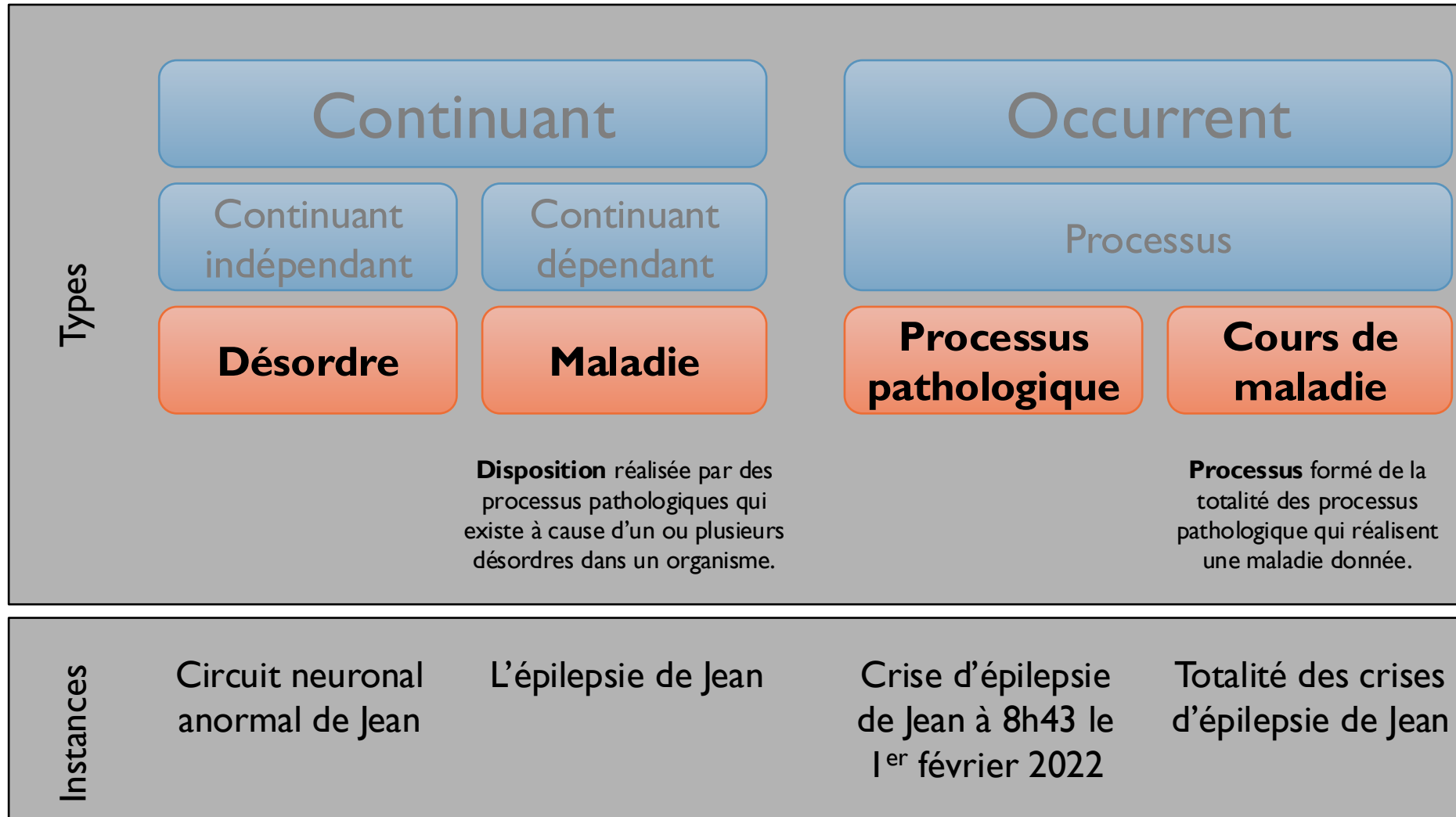


Application à la classification : l'exemple de la maladie



OGMS : Une ontologie générale de la maladie

Scheuermann, R. H., Ceusters, W., & Smith, B. (2009). Toward an ontological treatment of disease and diagnosis. *Summit on translational bioinformatics*, 2009, 116.



Cardiovascular Disease Ontology:

Quelques exemples de classes

Disorder

- ⊖ 'heart conduction disorder'
 - 'cardiac ion channels disorder'
 - 'impaired anterior division of left branch of atrioventricular bundle'
 - 'impaired atrioventricular node'
 - 'impaired posterior division of left branch of atrioventricular bundle'
 - 'impaired right branch of atrioventricular bundle'
 - 'impaired sinoatrial node'
 - 'impaired sinus node'

Disease

- ⊖ 'atrial fibrillation (disease)'
 - 'familial atrial fibrillation'
 - 'paroxysmal atrial fibrillation'
 - 'permanent atrial fibrillation'
 - 'persistent atrial fibrillation'
 - 'tachycardia-bradycardia syndrome'

Désambiguation

Pathological Process

- ⊖ 'myocardium pathological process'
 - ⊖ 'inflammation process in a myocardium'
 - ▶ ⊖ 'myocardial infarction'
 - ▶ ● 'myocardium contraction dysfunctional process'
 - 'myocardium of left ventricle stiffening'
 - ▼ ● 'ventricular myocardium dilatation'
 - ⊖ 'left ventricular myocardium dilatation'
 - ⊖ 'right ventricular myocardium dilatation'
 - ▼ ● 'ventricular myocardium hypertrophy'
 - ⊖ 'left ventricular myocardium hypertrophy'
 - ⊖ 'right ventricular myocardium hypertrophy'

- 'atrial fibrillation (process)'

Barton, A., Rosier, A. Burgun, A & Ethier, J.-F.
(2014) The Cardiovascular Disease Ontology.
In: *Proceedings of the 8th International Conference on
Formal Ontology in Information Systems (FOIS-
2014)*, Amsterdam: IOS Press, 409-414.

Les définitions en ontologie

Définitions en
langage courant
(aristotéliennes)

Définitions
logiques

Annotations: 'cardiomyopathy (1995 definition)'

'CVDO definition'

Disease of the myocardium associated with cardiac dysfunction.
(Report of the 1995 World Health Organization/International Society and Federation of Cardiology Task Force on the Definition and Classification of Cardiomyopathies)

Description: 'cardiomyopathy (1995 definition)'

Equivalent To +

● **disease**
and ('realized in' some
(**disease course**
and ('has occurrent part' some 'ventricular myocardium contraction dysfunctional process')))

Classé comme consistant par le raisonneur Pellet

Annotations: 'cor pulmonale'

'CVDO definition' [language: en]

A disease characterized by a dilatation or hypertrophy of the myocardium of right ventricle of the heart as a response to increased resistance or high blood pressure in the lungs, involving a failure of the right side of the heart.

Assertions et inférences

Description: 'cor pulmonale'

Equivalent To +

SubClass Of +

● **'realized in' some**
(**disease course**
and (('has occurrent part' some 'right ventricular myocardium dilatation')
or ('has occurrent part' some 'right ventricular myocardium hypertrophy'))
and ('has occurrent part' some 'right ventricular myocardium contraction dysfunctional process'))

● **disease**

→ 'cardiomyopathy (1995 definition)'

→ 'right ventricle disease'

Rend compte de l'évolution des définitions

● **'myocardium disease'**

→ ● **'cardiomyopathy (1995 definition)'**

→ ● **'cardiomyopathy (2008 definition)'**

▶ ● **'dilated cardiomyopathy'**

▶ ● **'genetic cardiomyopathy'**

▶ ● **'hypertrophic cardiomyopathy'**

▶ ● **'left ventricular noncompaction'**

▶ ● **'restrictive cardiomyopathy'**

▶ ● **'Tako Tsubo cardiomyopathy'**

→ ● **'cor pulmonale'**

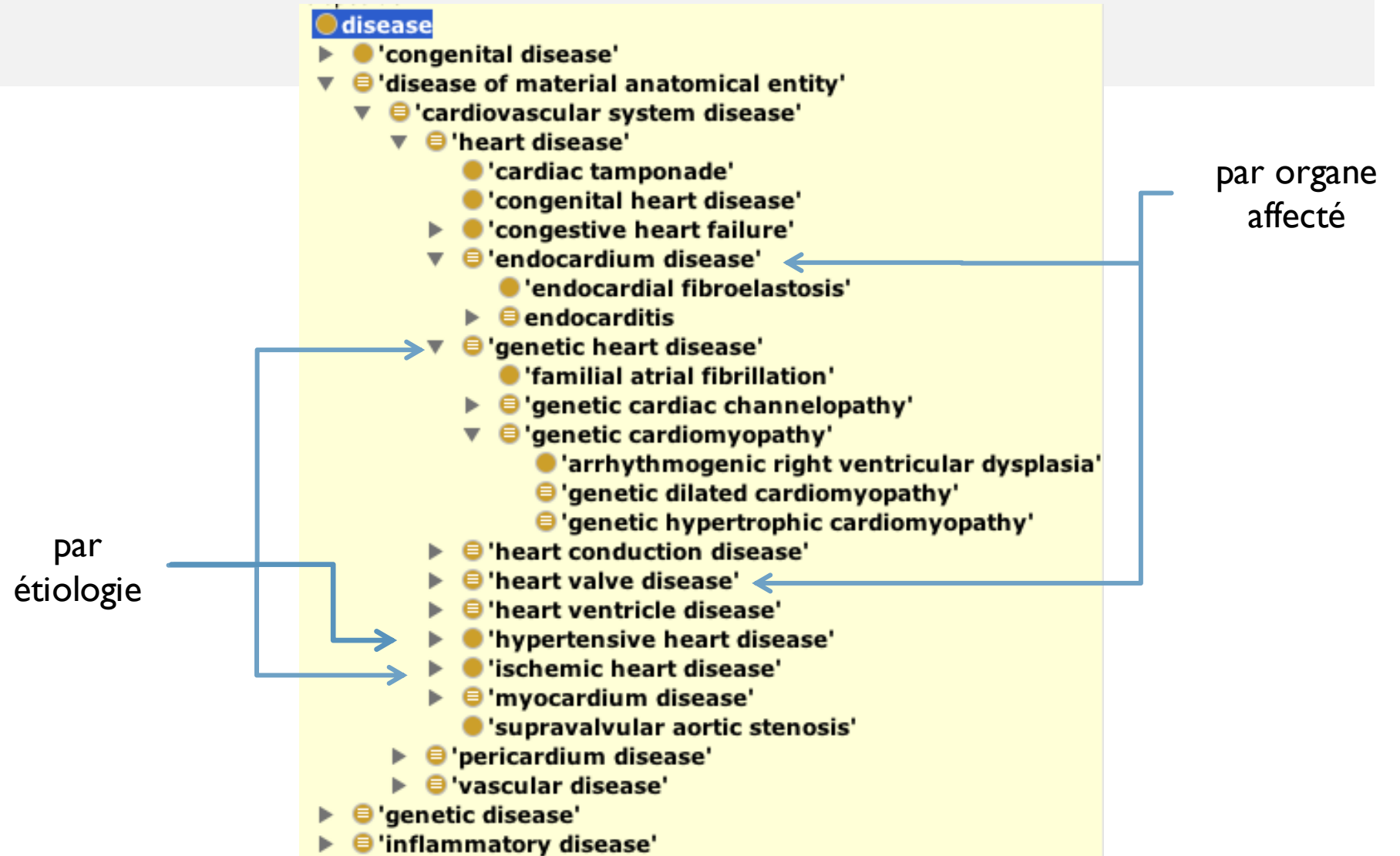
→ ● **'hypertensive cardiomyopathy'**

→ ● **'ischemic cardiomyopathy'**

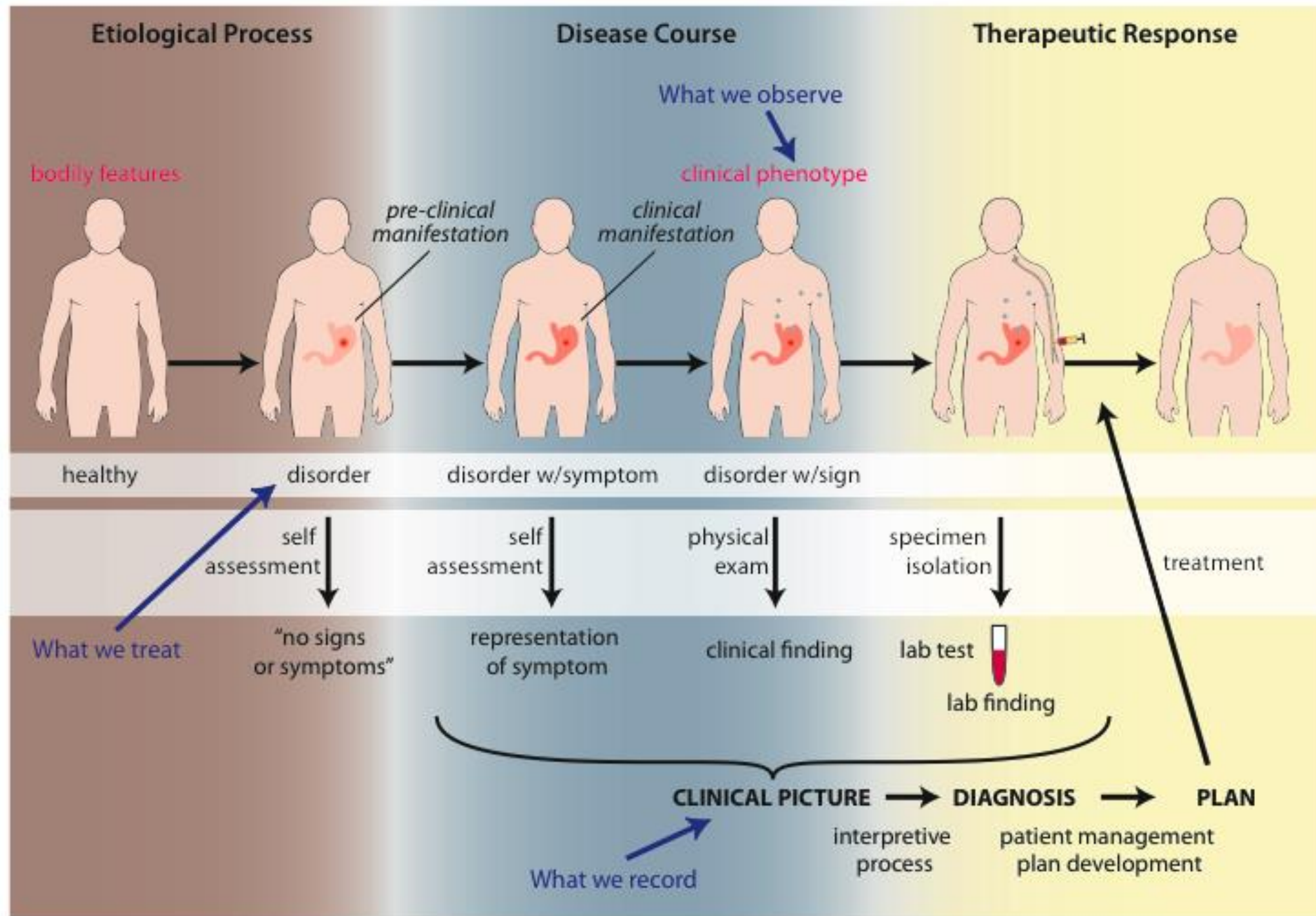
→ ● **'valvular cardiomyopathy'**

Révèlent des défauts logiques dans des
définitions courantes

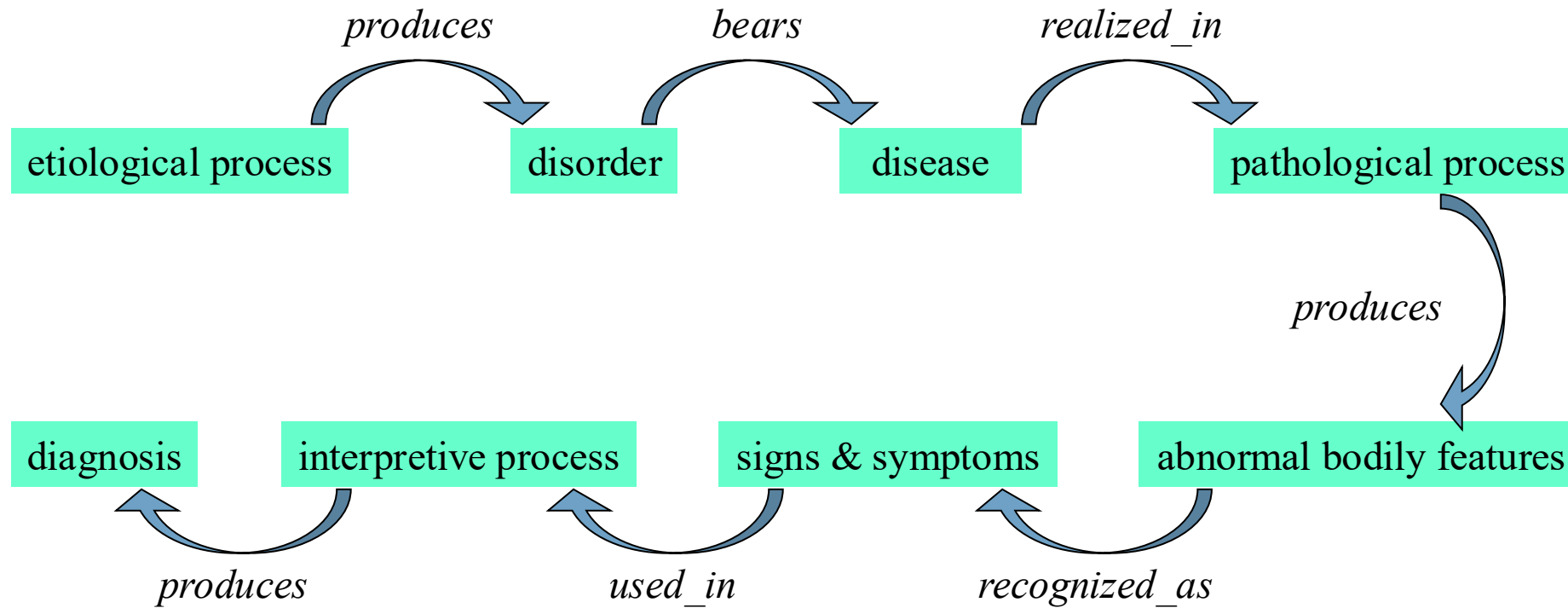
Classification automatique des maladies



Scheuermann, R. H., Ceusters, W., & Smith, B. (2009). Toward an ontological treatment of disease and diagnosis. *Summit on translational bioinformatics*, 2009, 116.



Aperçu de possible relations pertinentes

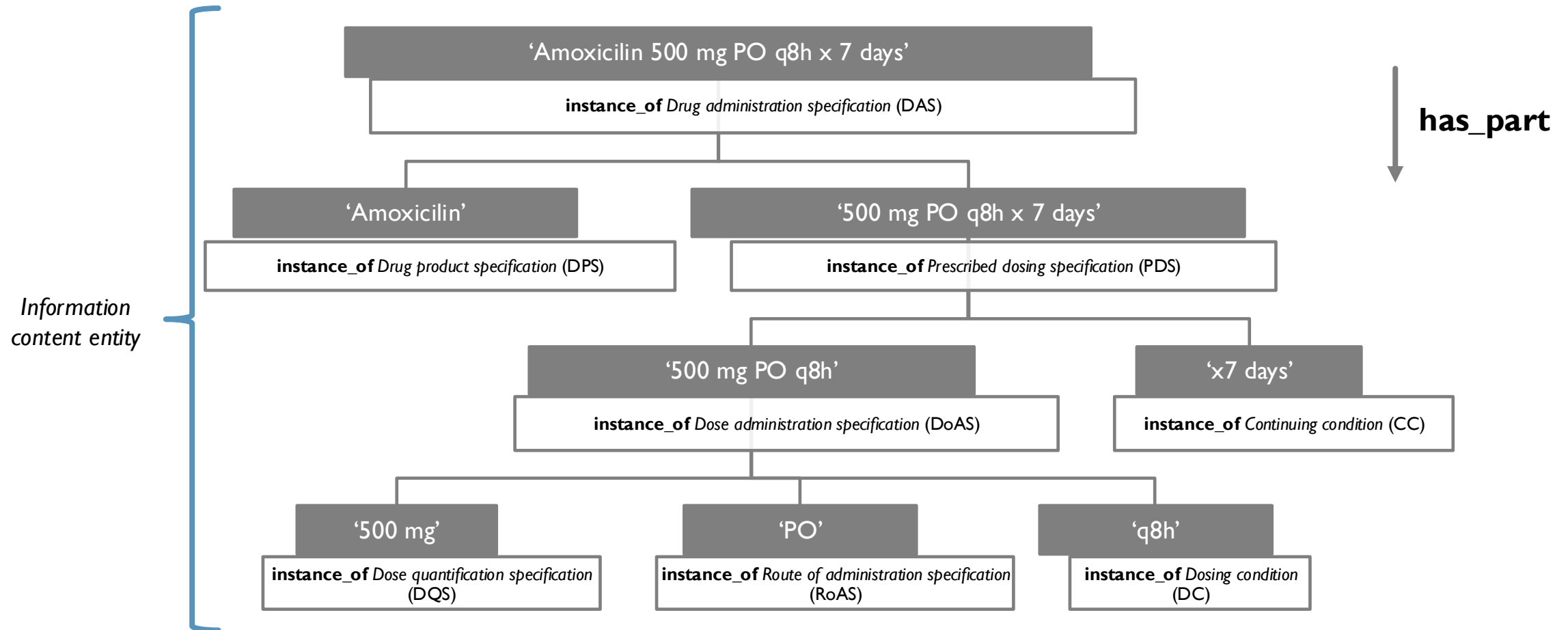


Influenza (infectious disease)

- Etiological process - infection of airway epithelial cells with influenza virus
 - *produces*
- Disorder - viable cells with influenza virus
 - *bears*
- Disposition (disease) - flu
 - *realized_in*
- Pathological process - acute inflammation
 - *produces*
- Abnormal bodily features
 - *recognized_as*
- Symptoms - weakness, dizziness
- Signs - fever
- Symptoms & Signs
 - *used_in*
- Interpretive process
 - *produces*
- Hypothesis - rule out influenza
 - *suggests*
- Laboratory tests
 - *produces*
- Test results - elevated serum antibody titers
 - *used_in*
- Interpretive process
 - *produces*
- Result - diagnosis that patient X has a disorder that bears the disease flu

PDR0: Prescription of Drugs Ontology (extrait)

Ethier, J. F., Goyer, F., Fabry, P., & Barton, A. (2021). The prescription of drug ontology 2.0 (PDR0): more than the sum of its parts. *International Journal of Environmental Research and Public Health*, 18(22), 12025.



Comment représenter formellement une ontologie ?

- Langages logiques :

- logique du premier ordre
- logique des description
- ...

$\forall x(\text{Univ}(x) \rightarrow \text{EducInst}(x)), \forall x(\text{Student}(x) \rightarrow \text{Person}(x)) \dots$

$\text{Univ} \sqsubseteq \text{EducInst}, \text{Student} \sqsubseteq \text{Person}, \text{Teacher} \sqsubseteq \text{Person}$

- Langages informatiques :

- RDF, RDFS
- OWL
- CLIF
- ...

Pizza SubClassOf
(has_topping min 1 Topping)



14:30 – 16:00
Ontologie des pizzas

Snape teaches Harry
Student SubClassOf Person



10:30 – 12:30
Introduction RDF, RDFS et SPARQL

Questions ?

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École d'été interdisciplinaire en numérique de la santé (EINS 2026)

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- François Goyer
- Olivier Grenier
- Christina Khnaïsser
- Luc Lavoie
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- ...

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- Cédric Tarbouriech (IRIT, Toulouse, France)
- Fumiaki Toyoshima (Université de Neuchâtel, Suisse)
- Laure Vieu (IRIT, Toulouse, France)
- ...